Smart Surplus Food Redistribution & Rescue Coordination Platform

A MINI PROJECT REPORT

IT2304 – FULL STACK WEB DEVELOPMENT

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECNOLOGY



DEPARTMENT OF INFORMATION TECHNOLOGY

KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY

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OCTOBER 2026

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BONAFIDE CERTIFICATE

Certified that the project report "**SMART SURPLUS FOOD REDISTRIBUTION & RESCUE COORDINATION PLATFORM"** is the bonafide work **of "SONA.S (9204242050100) & JANANI.J(920424205034) & PANDEESWARI.N (920424205071)"** who carried out the mini project work as a part of **IT2304 – Full Stack Web Development** under my supervision.

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CHAPTER 1

INTRODUCTION

1.1 MERN Stack

MERN Stack is a popular full-stack JavaScript technology stack used for developing modern web applications. MERN stands for MongoDB, Express.js, React.js, and Node.js. It provides technologies for developing the frontend, backend, and database of a web application. In this project, the MERN Stack is used to develop FoodRescue, a web-based platform that connects food donors, NGOs, and volunteers to manage surplus food redistribution efficiently.

1.2 Importance of MERN Stack

MERN Stack simplifies web application development by using JavaScript across different layers of the application. It helps developers build applications efficiently, reduce development complexity, and maintain a consistent programming language throughout the project. It supports dynamic user interfaces, database management, and communication between frontend and backend services. MERN Stack is suitable for developing scalable and interactive applications such as FoodRescue.

1.3 Basic Terminologies

- **MongoDB**: A NoSQL database used to store and manage application data in document format.
- **Express.js:** A lightweight web framework for Node.js used to develop backend applications and REST APIs.
- **React.js:** A JavaScript library used to create dynamic and interactive user interfaces.
- **Node.js:** A JavaScript runtime environment used to execute JavaScript on the server side.
- **Food Donation:** The process of sharing surplus food with organizations that can distribute it to people in need.
- **NGO:** A non-governmental organization that can accept suitable food donations and coordinate their distribution.

- **Volunteer Coordination:** The process of arranging volunteers to collect and deliver donated food.
- **Smart Matching:** A basic matching mechanism that recommends suitable NGOs or volunteers based on factors such as location, capacity, availability, and pickup deadline.

1.4 Components of MERN Stack

1.4.1 MongoDB

MongoDB is a NoSQL database that stores data in flexible document-based formats. In the FoodRescue project, it is used to store user details, food donation information, NGO details, volunteer information, and donation status. It helps manage and retrieve application data efficiently.

1.4.2 Express.js

Express.js is a backend framework built on Node.js. It is used to create server-side applications, handle HTTP requests, and develop REST APIs. In FoodRescue, Express.js manages user authentication, food donation operations, NGO acceptance, volunteer coordination, and communication between the frontend and database.

1.4.3 React.js

React.js is a JavaScript library used to develop dynamic and interactive user interfaces. It supports reusable components and helps build responsive web pages. In FoodRescue, React.js is used to develop the landing page, login and registration pages, and separate dashboards for donors, NGOs, volunteers, and administrators.

1.4.4 Node.js

Node.js is a JavaScript runtime environment that allows JavaScript code to run on the server. It is used with Express.js to build backend services and process client requests. In FoodRescue, Node.js supports application logic, API processing, authentication, and donation workflow management.

1.5 Operations in MERN Stack

- Create: Donors can add new food donation details to the platform.
- Read: Users can view available donations and relevant information based on their roles.

- Update: Donation details and statuses can be updated during acceptance, pickup, and delivery.
- Delete: Authorized users can remove eligible records when permitted by the application.
- API Communication: Data is transferred between the React frontend and Express/Node.js backend through APIs.
- Donation Tracking: The application tracks the progress of donations from availability to final delivery.

1.6 Need for MERN Stack

- Full-stack web development using JavaScript.
- Efficient storage and management of food donation data.
- Interactive dashboards for donors, NGOs, volunteers, and administrators.
- Easy communication between frontend and backend through REST APIs.
- Role-based access to application features.
- Flexible management of donation details and status updates.
- Support for developing and extending the application with additional features.

1.7 Applications of MERN Stack

- E-commerce applications.
- Online food ordering and delivery systems.
- Social media applications.
- Content management systems.
- Booking and reservation systems.
- Real-time web applications.
- Business and management applications.
- Food donation and redistribution platforms.

FoodRescue applies MERN Stack technology to connect food donors, NGOs, and volunteers through a centralized platform. It helps organize surplus food donations, coordinate pickup and delivery, and track donation progress. The system aims to reduce food wastage and improve the efficiency of food redistribution.

CHAPTER 2

METHODOLOGY

2.1 Objective

The main objective of the FoodRescue project is to develop a web-based platform for efficiently coordinating surplus food redistribution.

The specific objectives are:

1. To provide a platform for donors to register and share surplus food.
2. To allow NGOs to view and accept available food donations.
3. To connect volunteers with food pickup and delivery tasks.
4. To maintain food donation and delivery status digitally.
5. To reduce manual coordination between donors, NGOs, and volunteers.
6. To store donation and user information securely using MongoDB.
7. To provide role-based access for donors, NGOs, volunteers, and administrators.
8. To develop a user-friendly and responsive web application using the MERN Stack.
9. To provide basic smart matching based on factors such as location, capacity, and availability.
10. To improve the overall organization and transparency of the food redistribution process.

2.2 Problem Statement

Surplus food from restaurants, hotels, hostels, college canteens, and individuals is often wasted because there is no proper system to quickly connect food donors with organizations that can redistribute the food.

The existing process may involve phone calls, messages, or manual coordination between donors, NGOs, and volunteers. This can result in delays, lack of information, difficulty in tracking donations, and inefficient volunteer coordination.

Therefore, there is a need for a centralized digital platform that can:

- Connect food donors with NGOs.
- Display available surplus food.

- Manage donation information.
- Coordinate volunteers.
- Track pickup and delivery.
- Maintain donation status.
- Provide centralized data management.

2.3 Project Description

FoodRescue is a MERN Stack-based web application that connects food donors, NGOs, and volunteers to reduce surplus food wastage.

Donors can post food details, while NGOs accept suitable donations and volunteers manage food pickup and delivery with status tracking.

The system provides basic smart matching based on location, food quantity, NGO capacity, pickup deadlines, and volunteer availability.

Major Modules

1. Authentication Module

Provides registration, login, JWT authentication, and role-based access.

2. Donor Module

Allows donors to create, manage, and track surplus food donations.

3. NGO Module

Allows NGOs to view available donations, accept suitable food, and monitor deliveries.

4. Volunteer Module

Allows volunteers to view assigned tasks and update pickup and delivery status.

5. Admin Module

Allows administrators to manage users, NGOs, volunteers, and donations.

6. Donation Management Module

Maintains food details, quantity, location, deadline, and current donation status.

7. Smart Matching Module

Provides basic recommendations based on location, capacity, quantity, availability, and deadline.

CHAPTER 3

RESULTS AND DISCUSSION

The FoodRescue application was successfully designed as a MERN Stack web application to manage surplus food donation and redistribution. The system provides separate functionalities for donors, NGOs, volunteers, and administrators.

The developed application allows users to register and login based on their roles. Donors can create food donations, NGOs can view and accept available donations, and volunteers can manage pickup and delivery activities. The system stores the application data in MongoDB and communicates between the frontend and backend through REST APIs.

The major screens of the developed application are shown below

Screenshots

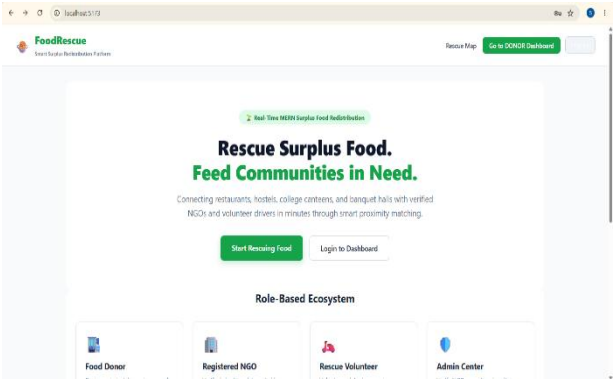


Figure 3.1 The landing page provides an introduction to the FoodRescue platform and allows users to access the login and registration pages.

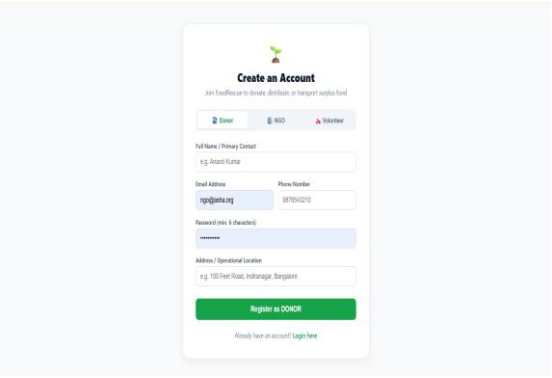


Figure 3.2 The registration page allows users to create an account by entering their basic details and selecting their appropriate role.

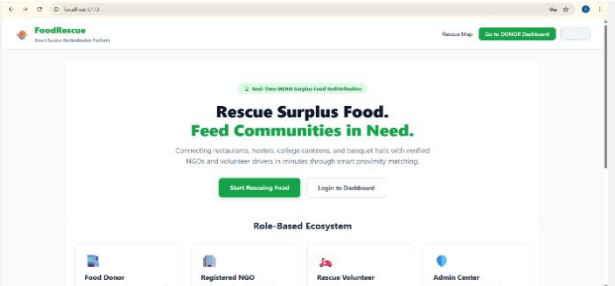


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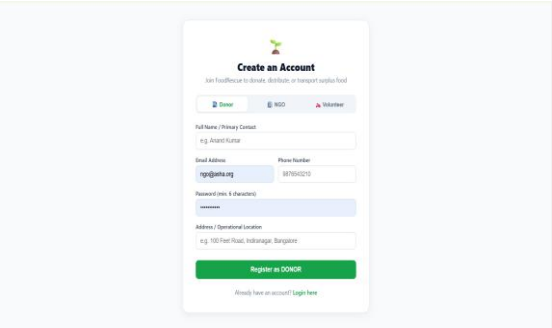


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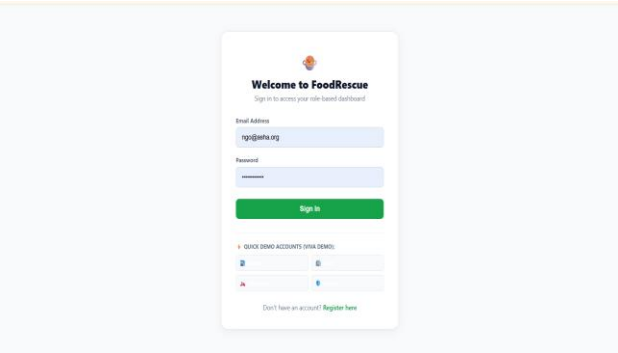


Figure 3.3 The login page authenticates registered users and redirects them to their respective dashboards based on their role.

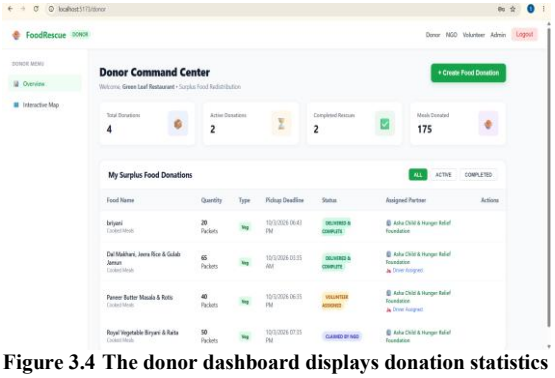


Figure 3.4 The donor dashboard displays donation statistics and provides options to create and manage surplus food donations.

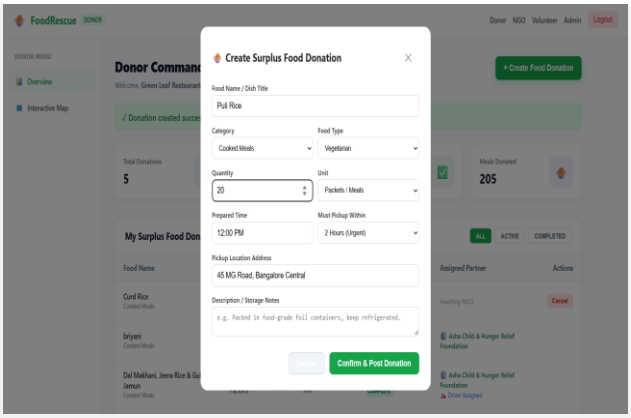


Figure 3.5 The donation form allows the donor to enter food name, category, quantity, food type, preparation time, pickup deadline, location, and description.

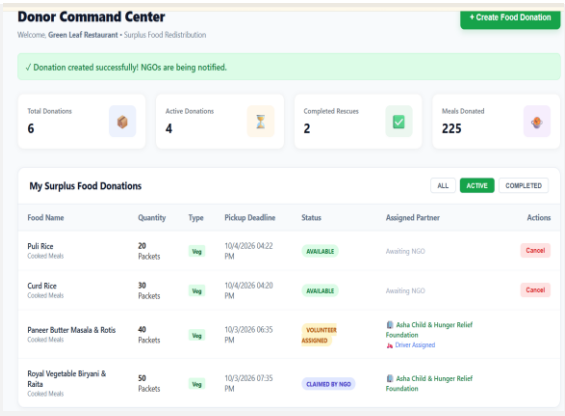


Figure 3.6 This page displays the donations created by the donor along with their current status.

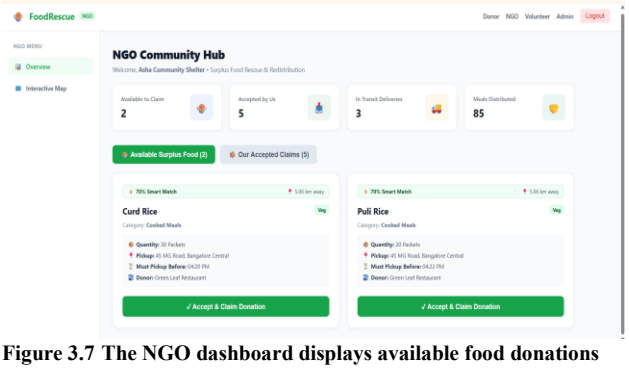


Figure 3.7 The NGO dashboard displays available food donations and accepted donations that can be managed by the organization.

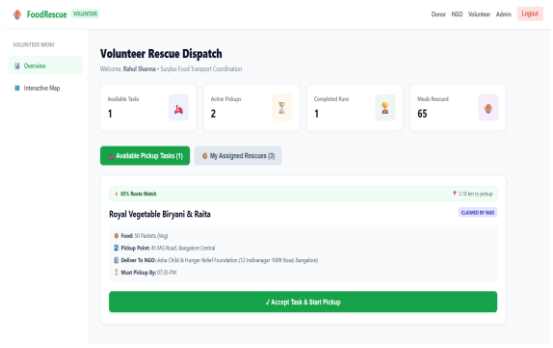


Figure 3.8 The volunteer dashboard displays assigned food pickup and delivery tasks along with their current status.

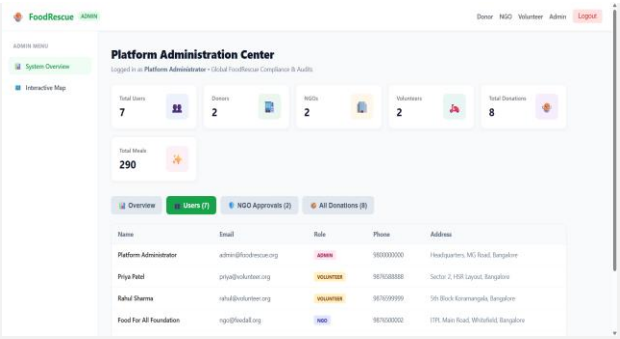


Figure 3.9 The admin dashboard provides an overview of users, donations, NGOs, volunteers, and overall system activity.

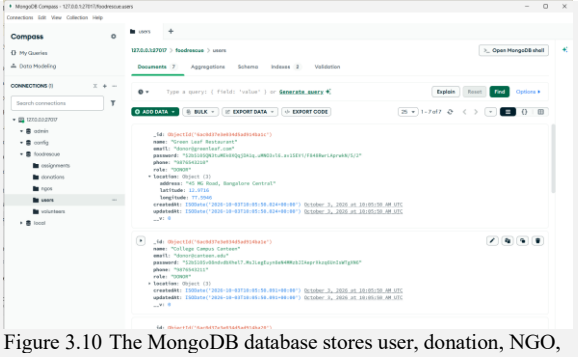


Figure 3.10 The MongoDB database stores user, donation, NGO, volunteer, and assignment information used by the application.

Github Link:

https://github.com/sonaeswari778607-hue/Smart-Surplus-Food-Redistribution-Rescue-Coordination-Platform

CHAPTER 4

CONCLUSION

The FoodRescue – Smart Surplus Food Redistribution & Rescue Coordination Platform was developed as a MERN Stack-based web application to provide an organized solution for managing surplus food redistribution.

The system connects donors, NGOs, and volunteers through a centralized platform. Donors can register surplus food, NGOs can accept available donations, and volunteers can coordinate pickup and delivery. The application also maintains donation status and stores the required information in MongoDB.

The project demonstrates the practical implementation of React.js, Node.js, Express.js, MongoDB, Mongoose, REST APIs, JWT authentication, and role-based authorization in a full-stack application.

FoodRescue can help reduce unnecessary food wastage by making the process of identifying, coordinating, and redistributing surplus food more systematic. The project can also be extended in the future with advanced features such as real-time tracking, QR-based verification, notifications, advanced matching, and analytics.

Overall, the project demonstrates how a MERN Stack web application can be used to address a real-world social problem through digital coordination and centralized information management.

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